Recommendation and Information Diffusion on Epinions

SNTP Assignment 2 — Technical Report

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Course	Social Network Theory and Programming (SNTP)
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Campus	Chiniot-Faisalabad Campus
Submission	20 October 2026
Dataset	Epinions ratings + trust network
Measured run	Bounded fast demo (full data files, 2-hop diffusion graph)

This report is generated from the saved JSON artifacts of the measured fast-demo run. It covers all eight requirements: data preparation, baseline recommenders, trust-aware recommendation, ICM/SIR diffusion, influence maximization, group recommendation, comprehensive evaluation, and the diffusion-limiting intervention with CELF and cost scaling. All cascade and satisfaction-after-intervention results are simulations; the dataset contains no adoption timestamps.

The measured fast-demo wall-clock runtime was 545.7 s (~9.1 min), including data loading, model fitting, all experiment stages, and result serialization. The larger full-scale configuration was not executed; no full-scale result is claimed.

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Source of reported numbers

Every numeric value in the tables is read from `results/fast_demo/*.json` at report-generation time. The stage timing table comes from `stage_timings.json`. The report is regenerated with the command shown in Section 11. This makes the PDF searchable and keeps the prose synchronized with the saved artifacts rather than with hand-copied values.

1. Executive summary and requirement map

The dataset contains 49,289 users, 139,738 items, 664,824 ratings, and 487,002 trust edges. The split is 463,663 train / 100,603 validation / 100,558 test ratings. Models are fitted on train; validation selects the social weight; test is used only for the final observed evaluation.

The strongest measured result is that matrix factorization and the trust-aware social model substantially outperform item-based CF on rating error. The social model is best on test MAE/RMSE and on all top-K ranking metrics in this run. Influence maximization does not show greedy superiority: on the restricted top-degree candidate pool, degree centrality wins at budgets 3 and 5, and the paired greedy−degree difference at budget 5 is negative. Group-strategy satisfaction differences are small, and the spread comparison is dominated by rare large cascades. The threshold intervention has a cliff at 0.8: it removes ~99% of simulated spread while slightly raising the simulator-side adopter rating proxy, but it is not an online experiment. CELF used 24.2% fewer evaluations and was 1.22× faster in this single noisy-Monte-Carlo measurement, while selecting different seeds; a deterministic controlled check reproduced standard greedy exactly.

Requirement	Implementation	Headline measured outcome
1. Data	Parse ratings/trust, validate, split, represent	49,289 users; 463,663/100,603/100,558 split
2. Recommenders	Item-based CF and MF; top-K	MF test MAE 0.8325, RMSE 1.0677
3. Trust-aware	Social blend; λ selected on validation	λ=0.3; test MAE 0.8284
4. Diffusion	ICM, SIR, preference-aware; controlled seeds	p=0.05 canonical 4,684 adopters; preference 501
5. Influence	Random/degree/greedy; shared pool and seeds	Degree best at budgets 3 and 5
6. Groups	Average / least misery / most pleasure; train-only groups	Mean satisfaction 4.827 / 4.825 / 4.815
7. Evaluation	MAE/RMSE/P/R/F1/HR/nDCG/coverage/diversity	Social best ranking; ICF broadest coverage
8. Intervention	Threshold, CELF, network/item/budget scaling	0.8: −99.09% spread, +0.0264 proxy; CELF 1.22×

Table 1. Requirement map and headline outcomes from the fast-demo run.

What is and is not claimed

- All diffusion, influence, group-spread, simulated-interaction, and intervention outcomes are simulations. The data has no adoption histories.
- The submitted results use a 2-hop ego subgraph for diffusion. They do not describe the full 49,289-user graph.
- No statistical significance is claimed for small differences. Monte Carlo standard deviations and sample scopes are reported.
- The full-scale config is documented but not run; the report makes no full-scale performance claim.

2. Dataset, split, and leakage protocol

2.1 Dataset

Property	Value
Users	49,289
Items	139,738
Ratings	664,824
Trust edges	487,002
Rating matrix sparsity	0.999903
Mean / median ratings per user	16.55 / 6.0
Mean / median ratings per item	4.76 / 1.0
Reciprocal trust edges	105,966
Users with both ratings and trust	40,162

Table 2. Parsed Epinions dataset summary.

Trust rows use the convention “A trusts B”. Influence propagates in the opposite direction: a propagation edge B → A means B can influence A. All parsing counters for malformed, out-of-range, and duplicate records are saved in `data_summary.json`; this run found zero malformed rating lines, zero out-of-range ratings, and zero duplicate rating records.

2.2 Split and evaluation discipline

The split seed is 42 with fractions 0.70/0.15/0.15. Users with at least three ratings contribute at least one record to each split; users with fewer records are kept for training but excluded from evaluation. The split sizes are 463,663 / 100,603 / 100,558. 28,487 users are eligible for evaluation and 11,676 are excluded.

2.3 Leakage prevention

- Model fitting: only training triples. Validation is used for the social λ; test is used only for final reported metrics.
- Group construction (R6): a matrix built explicitly from the training triples. The full `dataset.rating_matrix` is never passed into R6. A regression test mutates the full matrix and asserts that the groups do not change.
- Target item (R4/R7/R8): selected from a training-only popularity band, with the rule and counts saved in JSON.
- Seed sets (R4/R6/R8): generated once and reused across comparable settings. Common random numbers are used for paired comparisons.
- Observed vs simulated: test ratings are labels, not recommender inputs. Simulated diffusion uses predicted affinity, not held-out ratings.
- Candidate pools: all catalogue items except the user's training history for ranking; an explicit shared top-degree pool for R5. The restriction is documented, not hidden.

3. Requirement 1 — data preparation

The loader validates every rating line, builds contiguous user/item indices, a sparse rating matrix, and a directed propagation graph. The splitter enforces the per-user minimum and records the exact sizes. The inspection command is:

```
python -m sntp.cli inspect --data-dir Assignment
```

The item popularity distribution is extremely long-tailed: 99,715 items have only 1–5 training/test interactions in the item-popularity analysis, while 3,449 items have 21+. This is why top-K ranking metrics are low and why the diffusion target is deliberately drawn from the middle training popularity band rather than from a popular item.

4. Requirement 2 — baseline recommendation

4.1 Models

Item-based CF uses cosine similarity between item interaction vectors, at most 20 neighbours, and at least 2 co-rated items. Matrix factorization uses 10 latent factors, learning rate 0.005, L2 regularization 0.02, 6 epochs, and seed 42. Both are fitted on the training split.

Top-K excludes the user's training items and ranks the remaining catalogue. Ranking metrics use a fixed seeded sample of 500 eligible users; MAE/RMSE use all test ratings. Unobserved items are not treated as negative examples.

Model	MAE	RMSE	P@10	R@10	F1@10	HR@10	nDCG@10	Coverage
Item-based CF	1.0740	1.4519	0.0004	0.00233	0.00061	0.0040	0.00086	0.01481
Matrix factorization	0.8325	1.0677	0.0020	0.00485	0.00229	0.0200	0.00300	0.00029

Table 3. Test rating and top-10 ranking metrics. MF is much better on error; both have low absolute ranking metrics on this sparse catalogue.

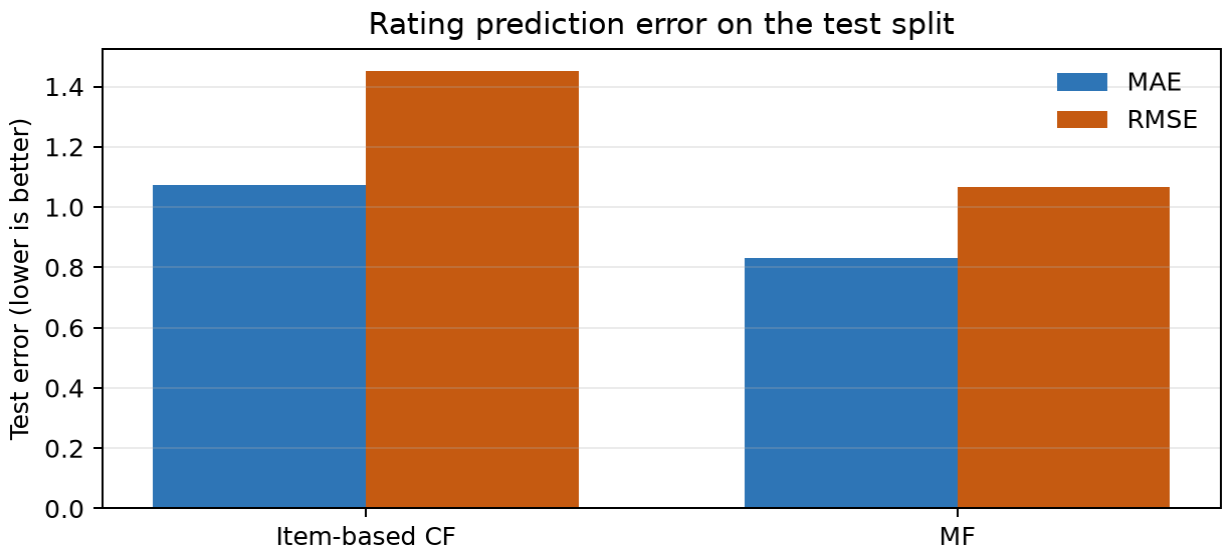


Figure 1. Test MAE and RMSE for the two baselines. MF reduces both errors substantially.

4.2 Behaviour and popularity slices

MF is consistently better than item-based CF on short, medium, and long user histories and on rare, moderate, and popular item groups. The largest gap is on moderate-popularity items: item-based CF MAE 1.2514 vs MF 0.8433. Item-based CF is not better on the rare-item group; its MAE is 0.9443 vs MF 0.8301. The complete slices are in `requirement_2_results.json`.

Interpretation: MF's low-rank structure handles sparse users and items better than a neighbour model with few co-rated items. The absolute top-K numbers are low because each user has many unobserved catalogue items and only one held-out positive item is used for relevance; this is a hard, conservative ranking setting, not evidence that the recommender is useless.

5. Requirement 3 — trust-aware recommendation

The social recommender blends the baseline prediction with the mean rating of trusted neighbours: `pred = (1−λ)·baseline + λ·trust_neighbour_mean`. The social weight is selected on validation RMSE only. The saved selection record explicitly states that test was not used.

λ	Validation MAE	Validation RMSE	N
0.0	0.83541	1.06899	100,603
0.3	0.83095	1.06397	100,603
0.5	0.83236	1.06934	100,603

Table 4. Validation selection. λ=0.3 has the lowest validation RMSE and is selected before test is examined.

λ	Test MAE	Test RMSE	P@10	HR@10	nDCG@10
0.0 (baseline)	0.83245	1.06772	0.0020	0.0200	0.00300
0.3 (selected)	0.82838	1.06319	0.0026	0.0240	0.00415
0.5	0.82998	1.06887	0.0030	0.0280	0.00481

Table 5. Test metrics reported after selection. λ=0.3 improves error and ranking relative to the baseline.

Sparse users (1–5 ratings, 21,911 users) improve from MAE 0.87453 to 0.87249 at λ=0.3. The trust-vs-similarity comparison sampled 50 users (165 test ratings): trusted neighbours MAE 0.87703, unconnected similar users 0.86647, baseline 0.87587. The similar-user group is slightly better in this small sample, so the trust signal is not uniformly superior; the result is descriptive.

The λ=0.5 test ranking is higher than λ=0.3 even though its validation RMSE is worse and its test RMSE is worse. This is exactly why the selection rule is fixed in advance: choosing λ=0.5 after seeing test ranking would be test leakage. The report keeps λ=0.3 and shows the post-selection λ=0.5 row for transparency.

6. Requirement 4 — ICM, SIR, and preference-aware diffusion

6.1 Design and target popularity

The target item is selected from the training-only rating-count distribution. The rule chooses the count closest to the median count in the inclusive [0.40, 0.60] training-count quantile band, with ties by ascending original item ID. The selected item is original ID 3 (matrix index 2), with 1 training rating. The band covers counts 1–2 and 83,981 items. It is intentionally a low-popularity item; absolute spreads are not a catalogue-wide forecast.

The three seed sets (five users each) and the Monte Carlo seeds are pre-generated once and reused for every ICM and SIR setting. The design therefore compares propagation models, not different random seeds:

```
[734, 3024, 4170, 7316, 13374]
```

```
[883, 1357, 2386, 2860, 13374]
```

```
[1413, 6945, 7018, 7985, 10792]
```

The fast demo uses 9,883 users and 301,710 propagation edges (2-hop ego subgraph centered on user 0), three simulations per setting, and at most 25 rounds. The SIR recovery probability γ=0 uses all 25 rounds because infected users never recover; those round counts are capped by the configured maximum, not evidence that the cascade naturally lasted 25 rounds.

6.2 Measured spreads

Model	Setting	Mean adopters/infected	Std	Mean rounds	Runtime (s)
Canonical ICM	p=0.01	125.33	167.36	9.00	0.066
Canonical ICM	p=0.05	4,684.33	21.70	11.00	0.702
Preference-aware ICM	p=0.01	6.00	0.82	1.00	0.005
Preference-aware ICM	p=0.05	501.33	351.96	7.67	0.200
SIR	beta=0.01_gamma=0.0	2,024.00	915.16	25.00	5.662
SIR	beta=0.01_gamma=0.2	530.67	692.39	17.67	1.396
SIR	beta=0.05_gamma=0.0	8,182.00	116.61	25.00	8.525
SIR	beta=0.05_gamma=0.2	6,021.00	20.54	25.00	4.646

Table 6. Diffusion outcomes. Means and standard deviations are across the same three pre-generated seed sets and controlled simulation seeds.

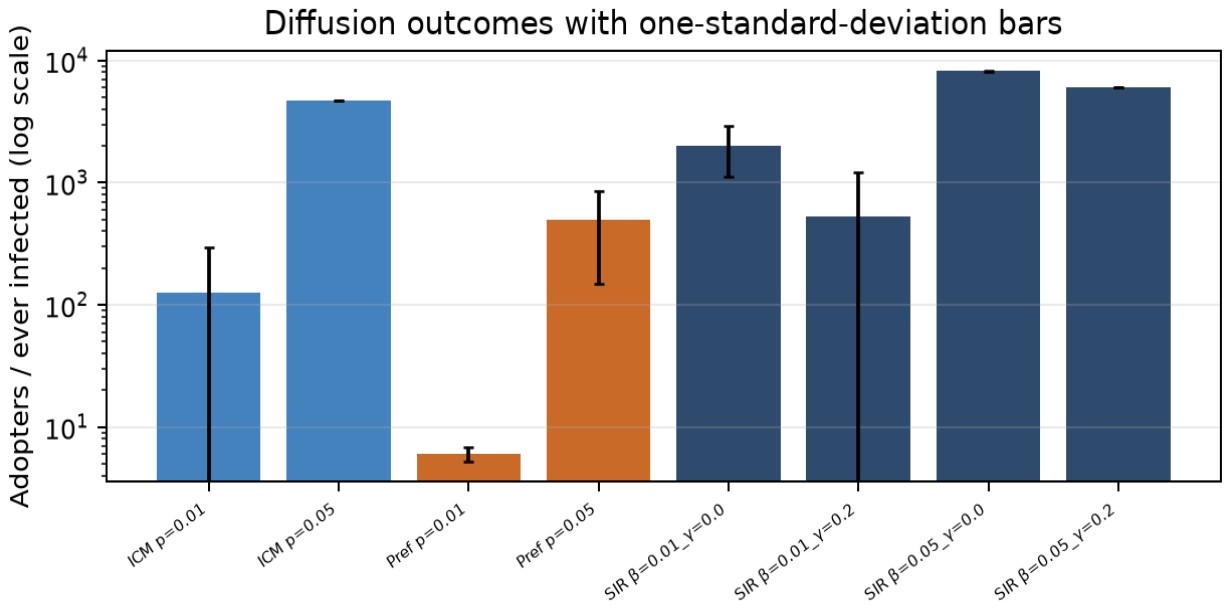


Figure 2. Diffusion outcomes on a log scale. The p=0.01 canonical ICM is extremely heavy-tailed; preference-aware adoption strongly suppresses spread at the same seeds.

6.3 Interpretation

Canonical ICM at p=0.05 reaches about 4,684 of 9,883 active users (47%), consistent with a dense ego graph. Preference-aware adoption requires predicted affinity and cuts the p=0.05 mean to 501 (−89.3%); at p=0.01 it cuts the mean from 125.3 to 6.0 (−95.2%). This is a model-design effect, not evidence that real users would adopt at that rate. SIR at β=0.05 reaches 8,182 with γ=0 and 6,021 with γ=0.2; recovery reduces ever-infected counts but the γ=0 runs remain active to the round cap.

The p=0.01 canonical ICM has mean 125.3 but standard deviation 167.4 and a maximum of 362: one simulation found a large cascade and two did not. With only three simulations, the mean is unstable. This is reported explicitly rather than hidden; the full configuration uses 20 simulations.

7. Requirement 5 — influence maximization

The comparison uses one propagation graph (9,883 nodes, 301,710 edges), one propagation probability (0.05), budgets 1/3/5, and a shared candidate pool. The pool is the top 15 out-degree users in the propagation graph (ties by ascending user index), identical for all methods. This bound is necessary for a fast demo, but it gives degree centrality a structural advantage. The result describes that pool, not all graph users.

Random selection is repeated for three seeded trials. Degree and greedy are deterministic selectors. Every method is evaluated on the same two evaluation seeds; random's mean is the mean of its trial means. Marginal gains are paired differences on the same Monte Carlo trials, not differences of unrelated runs.

Budget	Random mean	Degree mean	Greedy mean	Best measured
1	4,711.00	4,712.00	4,726.50	greedy
3	4,690.00	4,747.50	4,687.00	degree
5	4,692.50	4,712.50	4,700.50	degree

Table 7. Mean simulated spread under common evaluation seeds. The restricted top-degree pool favors degree; greedy is best only at budget 1.

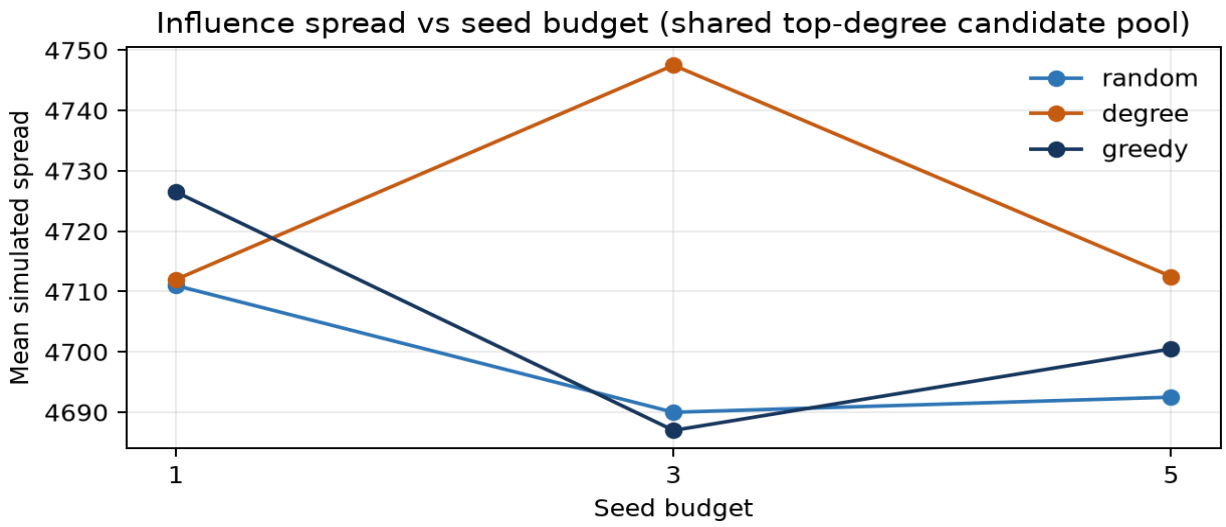


Figure 3. Greedy, degree, and random spreads. Differences are small relative to the saturated baseline and do not show greedy superiority.

7.1 Marginal gains and uncertainty

Method	Budget	Paired marginal gain	95% CI	Wins/losses
random	3	-21.00	—	—/—
random	5	2.50	—	—/—
degree	3	35.50	[26.68, 44.32]	2/0
degree	5	-35.00	[-60.48, -9.52]	0/2
greedy	3	-39.50	[-40.48, -38.52]	0/2
greedy	5	13.50	[-7.08, 34.08]	2/0

Table 8. Marginal gains relative to the previous evaluated budget. Negative values are legitimate under finite Monte Carlo sampling; they are not clipped to zero.

At budget 5, greedy − degree = −12.0 adopters (paired 95% CI [−19.84, −4.16]), and greedy loses both paired trials. Greedy exceeds random by 8.0 adopters (0.2%), far below the trial variability. The honest conclusion is no evidence of greedy superiority on this restricted pool. Degree and greedy adopter sets overlap substantially (mean Jaccard 0.551), so high-degree seeds largely reach the same audience.

Greedy selected the full budget in every row (65 evaluations at budget 5) and took 34.04 s for selection. Degree selection was effectively instant (0.0001 s). The standard-greedy sentinel is −∞, so a negative Monte Carlo gain does not stop seeding; this was a defect in the earlier implementation.

8. Requirement 6 — group recommendation

Groups are constructed only from the training split via `build_rating_matrix(train_ratings)`. The full observed matrix is never passed into R6; a regression test mutates the full matrix and asserts that the groups are unchanged. The construction forms conflicting groups from low, middle, and high mean training ratings, with seed 42. The fast demo uses 3 groups × 4 members; groups may overlap across groups.

The three aggregation strategies are average predicted rating, least misery (minimum), and most pleasure (maximum). Satisfaction is the recommender's prediction for each member, not a held-out rating. Diffusion uses preference-aware ICM with the group members as seeds, two simulations per candidate item, and controlled seeds.

Strategy	Mean satisfaction	Mean min	Mean range	Mean spread	Spread std
average	4.827	4.656	0.294	198.833	194.833
least misery	4.825	4.656	0.292	203.500	199.500
most pleasure	4.815	4.633	0.317	198.833	194.833

Table 9. Group strategy outcomes across three groups. Satisfaction differences are small; spread variability is large.

The average strategy has the highest mean and minimum satisfaction, while least misery has the highest mean simulated spread. The differences are tiny: mean satisfaction 4.827 vs 4.825 vs 4.815, and minimum satisfaction 4.656 vs 4.656 vs 4.633. The spread standard deviation (~195–200) is much larger than the between-strategy mean difference (~5). One group's cascade was 4 adopters in one simulation and ~1,171–1,199 in the other, depending on the seed; that single path dominates the average. These data do not support a strong “group strategy trade-off” claim.

8.1 Item-level comparison and ties

For each group, the candidate items are the union of the three strategies' top items, not the full catalogue. The item with the highest predicted mean satisfaction was also one of the highest-spread items in 2 of 3 groups (66.7%). Ties are treated as tied bests: in group 0 the best-spread set is {127, 16353}, so the satisfaction winner 16353 is a spread winner even though 127 is the canonical lowest-ID representative. In group 1 the satisfaction winner (16353) and spread winner (8559) differ. This tie-aware overlap is a non-empty intersection; it does not require the two best sets to be identical.

9. Requirement 7 — comprehensive evaluation

9.1 Metric definitions

- Relevance: held-out test rating ≥ 4. Unobserved items are not negatives.
- Candidates: all catalogue items except the user's training items; no negative sampling.
- Precision@10, Recall@10, HitRate@10, nDCG@10: standard per-user definitions, averaged over users. A user with no relevant held-out item contributes defined zeros for the ranking average.
- F1@10: mean of per-user F1 values, not the harmonic mean of aggregate precision and recall.
- Coverage: unique recommended items / full 139,738-item catalogue over the evaluated user sample. It is a sample-scope coverage, not a platform-wide estimate.
- Diversity: mean pairwise 1−cosine of training-interaction item feature vectors within each list, on a bounded deterministic list sample. It is feature diversity, not category diversity.
- No accuracy classification metric is reported: with only held-out positives and no observed negatives, a confusion-matrix accuracy would be misleading.

9.2 Observed evaluation

Model	MAE	RMSE	P@10	R@10	F1@10	HR@10	nDCG@10	Coverage	Diversity
Item-based CF	1.0740	1.4519	0.0004	0.00233	0.00061	0.0040	0.00086	0.01481	0.8590
MF	0.8325	1.0677	0.0020	0.00485	0.00229	0.0200	0.00300	0.00029	0.9856
Social (λ=0.3)	0.8284	1.0632	0.0026	0.00607	0.00307	0.0240	0.00415	0.00068	0.9848

Table 10. Observed test metrics (500 seeded users for ranking; all test ratings for MAE/RMSE). Social is best on error and ranking; ICF has the broadest coverage.

The social model is best on MAE, RMSE, precision, recall, F1, hit rate, and nDCG. Item-based CF has much higher coverage but its ranking accuracy is far lower. MF has very low coverage because its top-10 lists concentrate on a small set of high-scoring items; this is a known coverage/accuracy trade-off. The diversity values are high for MF/social (≈0.985) because their lists mix items with different interaction vectors, while ICF is lower (0.859).

9.3 Separate simulated interaction

The simulated interaction uses one training-selected target item and common seeds; it is not an observed adoption forecast. Mean adopters are ICF 753.0, MF 789.0, social 850.0. The highest simulated spread and the highest observed nDCG agree on the social model in this run, but the two quantities answer different questions. The report does not merge them into one score.

10. Requirement 8 — intervention, CELF, and scaling

10.1 Diffusion-limiting intervention

The intervention applies a minimum predicted-affinity threshold inside preference-aware ICM: exposed users below the threshold neither adopt nor propagate. Seeds are platform-promoted users and always start the cascade. The baseline is threshold 0.0 with the same seed sets and seeds as each threshold row. Five simulations are used per threshold.

Threshold	Mean adopters	Mean exposed	Conversion	Rating proxy	Spread change	Proxy change
0.0	771.00	1,984.00	0.3886	4.0774	—	—
0.3	771.00	1,984.00	0.3886	4.0774	0.00%	0.0000
0.6	772.20	2,030.80	0.3802	4.0921	+0.16%	0.0147
0.8	7.00	31.00	0.2258	4.1038	-99.09%	0.0264
0.9	5.20	17.20	0.3023	4.0489	-99.33%	-0.0285

Table 11. Intervention results relative to threshold 0. The rating proxy is 1+4×predicted affinity and is not an observed satisfaction rating.

The threshold effect is a cliff, not a smooth trade-off. The baseline exposure population is 4,850 unique users (across five simulations); affinity ranges 0.4434–0.9838. The share below threshold is 0.00% at 0.3, 0.72% at 0.6, 80.93% at 0.8, and 99.38% at 0.9. Threshold 0.3 therefore cannot change the baseline cascade; 0.6 changes only a few paths; 0.8 removes most propagation; 0.9 removes slightly more spread but lowers the proxy because the remaining adopters are mostly seeds. Shared seeds reduce but do not eliminate path divergence when filtering changes the exposed set.

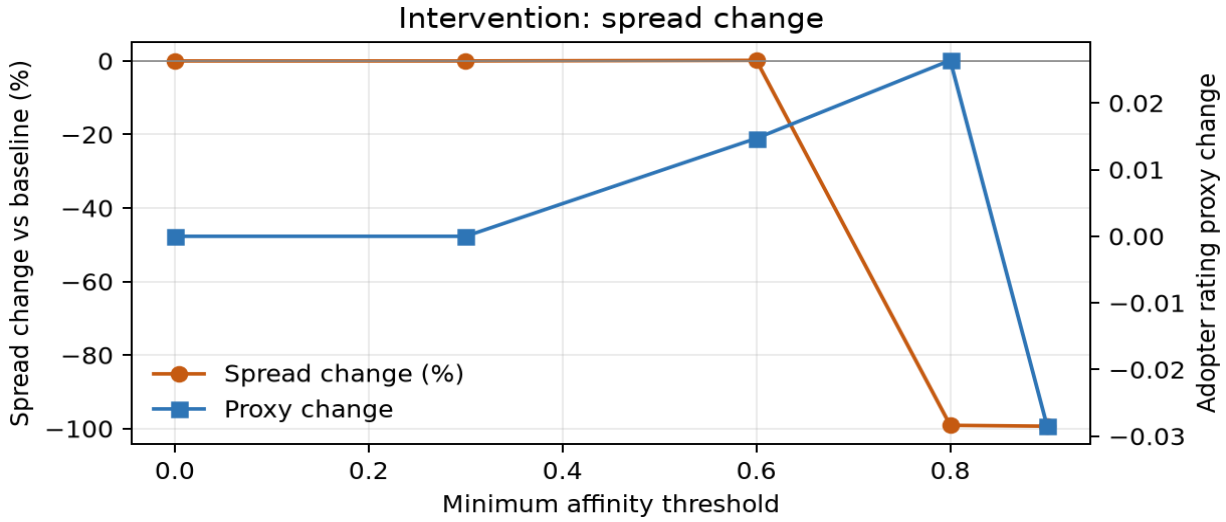


Figure 4. The 0.3 threshold is inactive; 0.6 is near baseline; 0.8 collapses spread and slightly raises the proxy; 0.9 lowers the proxy.

The honest reading: this intervention can strongly limit simulated diffusion at a high threshold, but the effect is not monotone and the proxy is not observed post-intervention satisfaction. The 0.8 row is the largest measured spread reduction among thresholds with a non-negative proxy change; the 0.9 row shows that over-filtering can also reduce the proxy. No online randomized experiment was run.

10.2 CELF optimization

Method	Evaluations	Runtime (s)	Selected seeds	Marginal gains
Standard greedy	33	83.45	22, 321, 570	4760.5, 0.5, 9.5
CELF	25	68.17	22, 321, 206	4760.5, 0.5, -13.5

Table 12. CELF vs standard greedy at budget 3, 12 candidates, 2 simulations. CELF used 24.24% fewer evaluations and was 1.224× faster, but selected different seeds under noisy estimates.

CELF's lazy bound is exact for a deterministic monotone submodular objective, not for a finite-sample noisy estimate. The measured 1.22× speedup and 24.2% evaluation reduction are therefore reported as single-run measurements, not an unconditional guarantee. A controlled deterministic check (p=1.0, one simulation, 120 nodes, 20 candidates, budget 2) gave 39 evaluations for both methods and identical seeds 494, 158. That check is the implementation correctness evidence; the noisy run is the cost evidence. Memory tracemalloc peaks were 8.452 MB and 8.453 MB; RSS deltas were 1.410 MB and 0.016 MB. NumPy native allocations may be under-represented.

10.3 Measured scaling and bottleneck

Network nodes	Edges	Runtime (s)	Evaluations	Tracemalloc peak (MB)
2,471	78,115	10.05	29	2.35
4,942	227,570	23.86	29	5.36

Table 13. Network-size scaling for greedy selection from scratch on BFS-induced subgraphs.

Catalogue items	ms/user	Runtime (s)	Tracemalloc peak (MB)
34,934	2.202	0.033	1.07
69,869	2.949	0.044	2.13
139,738	9.326	0.140	4.26

Table 14. Item-catalogue scaling for MF scoring/ranking. This isolates scoring, not model training.

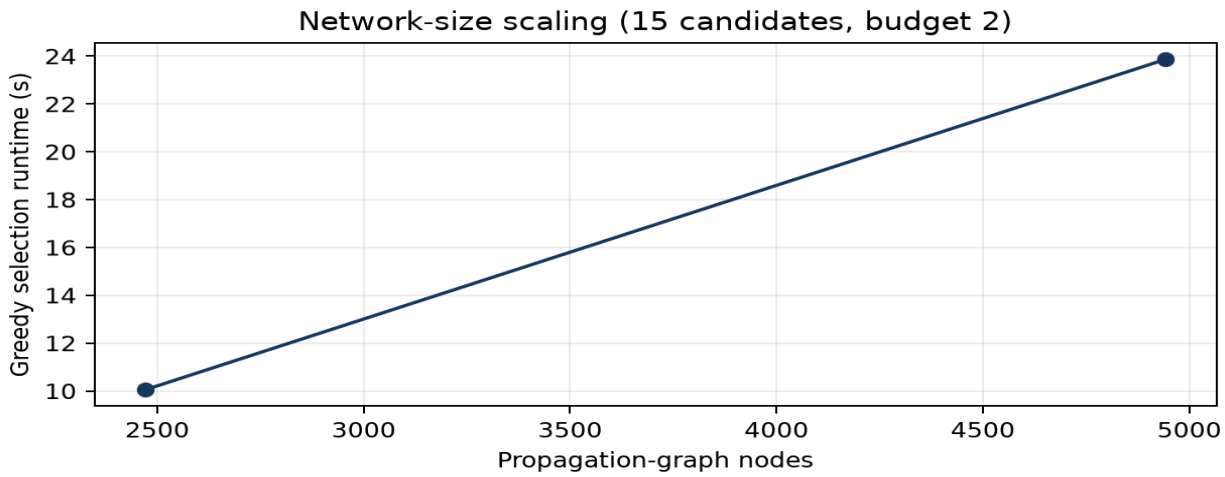


Figure 5. Measured network scaling. The slope is machine-dependent; the report compares relative growth within this run.

Seed budget	Evaluations	Runtime (s)
1	15	5.61
3	42	15.58

Table 15. Promotion-budget scaling on a 2,471-node subgraph. Evaluations grow with budget because greedy re-evaluates candidates.

Within R8, the CELF comparison is the dominant cost (151.62 s), followed by network scaling (33.91 s) and budget scaling (21.19 s). The intervention itself is cheap (1.83 s); the cost is repeated Monte Carlo evaluation, not the threshold arithmetic. Network runtime grows from 10.05 s at 2,471 nodes to 23.86 s at 4,942 nodes. Budget runtime grows from 5.61 s to 15.58 s as evaluations grow from 15 to 42. Item scoring grows from 2.20 to 9.33 ms/user as the catalogue grows from 34,934 to 139,738; this is a measured cache/allocation effect, not a claimed linear law.

11. Verification, tests, and reproducibility

11.1 Automated tests

The latest local run was 87 passed with `python -m pytest -q`. The leakage/fairness regressions are explicit:

- Training-only matrix construction and R6 groups unaffected by mutating the full observed matrix.
- Social λ selected on validation when test ratings are perturbed.
- Target-item selection and seed-set generation deterministic and training-only.
- Greedy continues after a negative Monte Carlo marginal gain; explicit candidate pool respected.
- Paired marginal gains equal the difference of the paired spread estimates.
- Feature diversity, per-user F1, and hand-checkable metric examples.

11.2 Exact commands

```
python -m pytest -q
```

```
python -m sntp.cli inspect --data-dir Assignment
```

```
python -m sntp.cli run --config configs/fast_demo.yaml
```

```
python -m pip install -r requirements-report.txt
```

```
python scripts/generate_report.py --results results/fast_demo --output report/SNTP_Assignment_02_Report.pdf --roll-number --student "Ahmad Fareed"
```

11.3 Runtime and seeds

Stage	Seconds
R1: load dataset	11.54
R1: train/val/test split	0.85
R2: item-based CF fit	87.84
R2: item-based CF MAE/RMSE	0.43
R2: item-based CF top-K ranking	1.42
R2: MF training	83.69
R2: MF MAE/RMSE	0.22
R2: MF top-K ranking	3.95
R2: behaviour/popularity group analysis	2.44
R3: baseline evaluation	5.16
R3: validation lambda selection	6.38
R3: social lambda=0.0	8.44
R3: social lambda=0.3	9.09
R3: social lambda=0.5	8.70
R3: sparse-user analysis	1.64
R3: trust vs similar-user comparison	5.09
R4: ICM canonical	0.77
R4: ICM preference-aware	0.21
R4: SIR	20.23
R5: Influence Maximization	42.12
R6: Group Recommendation	4.46
R7: Comprehensive Evaluation	19.38
R8: Intervention & Cost Analysis	212.73

Table 16. Stage timings from the submitted run. Total CLI wall clock: 545.7 s.

Config: `configs/fast_demo.yaml` (resolved copy in `results/fast_demo/experiment_config.json`). Split seed 42; ranking/diversity sample seed 42; MF seed 42; group seed 42; R5 selection seeds [49, 1049], evaluation seeds [500049, 501049]; R6 per-group seeds [100049, 101049], [101049, 102049], [102049, 103049]; R8 intervention seeds [900049, 901049, 902049, 903049, 904049]; R8 CELF selection seeds [800049, 801049]; scaling offset 820000. The

JSON files record the seed schedules used.

12. Limitations and conclusions

12.1 Limitations

- Full-scale run not executed. The full config uses the full graph, 20 simulations, larger candidate pools, and larger budgets; estimated cost is many hours. No full-scale number is claimed.
- Fast-demo scope. Diffusion/influence/group/intervention use a 2-hop ego subgraph (9,883 users); R5 uses a top-15 out-degree pool. The demo also uses smaller model budgets than the full config (ICF 20 neighbours vs 50; MF 6 epochs vs 20). These are bounded, deterministic scopes, not random full-graph samples.
- Monte Carlo noise. R4 uses 3 simulations; R5 uses 2 evaluation simulations and 3 random trials; R6 uses 2 per item; R8 CELF uses 2. Heavy tails make small mean differences unstable.
- R5 does not show greedy superiority. Degree wins at budgets 3 and 5; the paired greedy−degree difference at budget 5 is negative. The dense graph is near saturation (~47% of active users), compressing differences.
- R6 differences are small. Satisfaction ranges 4.815–4.827; spread is dominated by one rare cascade. Candidate items are only the union of strategy top items.
- R7 is sample-based. Ranking uses 500 users, coverage/diversity are list-sample properties, and unobserved items are not negatives. The simulation uses one low-popularity target item.
- R8 is a simulator intervention. No online experiment; the rating proxy is predicted, not observed; the threshold cliff is a property of the affinity distribution; CELF's 1.22× is a single noisy measurement and seeds differ. The deterministic check is the correctness evidence.
- Memory measurement. tracemalloc undercounts NumPy native buffers; RSS deltas are near zero because arrays are already resident.
- Runtime variability. Prior runs of the same fast config ranged 365–596 s; the saved stage timings are one run and should not be treated as a benchmark.

12.2 Conclusions by requirement

Requirement	Conclusion
Data/split	The parser is clean (zero malformed/duplicate/out-of-range ratings) and the 70/15/15 split is reproducible with per-user coverage.
Baselines	MF clearly beats item-based CF on rating error; both have low absolute top-K metrics on this sparse catalogue. MF is better across user-history and item-popularity slices.
Trust-aware	Validation-selected λ=0.3 gives a small but consistent test improvement (MAE 0.83245→0.82838) and better ranking. The trust-vs-similarity comparison is small and descriptive.
Diffusion	Canonical ICM at p=0.05 saturates the ego graph; preference-aware adoption reduces spread by ~89–95% at the same seeds. SIR recovery reduces ever-infected counts. Results are simulation-only and heavy-tailed at low p.
Influence	On a shared top-degree pool, degree is best at budgets 3 and 5; greedy is best only at budget 1. There is no evidence for greedy superiority, and the pool restriction is material.
Groups	Average has the best predicted mean/min satisfaction; least misery has the largest (but noisy) simulated spread. Differences are small; one rare cascade dominates spread.
Evaluation	Social is best on observed error and ranking; ICF gives the broadest coverage; MF gives the highest feature diversity. Observed and simulated outcomes are reported separately.
Intervention/optimization	Threshold 0.8 reduces simulated spread by 99.09% with a +0.0264 proxy change; 0.9 reduces spread slightly more but lowers the proxy. CELF uses 24.2% fewer evaluations and is 1.22× faster in the noisy run, with different seeds; a deterministic check matches standard greedy exactly.

Table 17. Conclusions tied to measured evidence. No unexecuted configuration is presented as a result.

The codebase is runnable, the eight requirements are implemented, and the submitted numbers are reproducible from the saved configuration and JSON. The main scientific caveats are the bounded fast-demo scope, the small Monte Carlo budgets, and the absence of observed adoption/satisfaction outcomes. Those caveats are reflected in the conclusions rather than hidden.